

Mohammed Shan Panikkaveetil Shoukathali

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PROFESSIONAL SUMMARY

Software Engineer with 2+ years of experience designing and building full-stack web applications, primarily using JavaScript/TypeScript, React, Node.js, and MongoDB, including production experience at Visa on large-scale systems. Comfortable working across the stack, from database design and RESTful API development to responsive, component-based front-end interfaces, with additional experience in Python, Java, and C++. MSc in Cyber Security (Distinction), with practical grounding in secure coding, authentication, and authorisation practices. Full right to work in the UK, open to relocation, seeking Software Engineer roles.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, Java, C++, SQL
- **Frontend:** React.js, Redux, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Vite
- **Backend:** Node.js, Express.js, RESTful API Design, WebSockets, Zod
- **Databases:** MongoDB (NoSQL), PostgreSQL, MySQL, Redis, Prisma, Database Design
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, Linux, Postman, npm, CI/CD basics
- **Testing:** Jest, TestContainers, Integration Testing
- **Security:** Secure Coding Practices, OWASP Top 10, Authentication & Authorisation, API Security

WORK EXPERIENCE

Software Engineer – Freelance, Remote (India) (Sep 2023 – Nov 2024)

- Developed and enhanced an admin dashboard and core platform features for a sports coaching marketplace connecting athletes and casual trainees with coaches, using React, Node.js, and MongoDB
- Developed a FastAPI and React-based ERP system for a vendor and inventory management platform in the aviation parts supply chain, handling vendor data, records, and workflow tracking
- Built a MERN-based student record and management system for a school client, covering student data, academic records, and administrative workflows
- Designed and implemented RESTful APIs with authentication and authorisation across all client projects
- Delivered responsive, component-based front-end interfaces across projects, ensuring consistent UX across devices

Software Engineer – Visa, Bangalore, India (Jun 2022 – Aug 2023)

- Rolled out service and configuration changes across production systems as part of a large-scale data centre migration, coordinating with cross-functional teams to validate deployments and maintain system reliability
- Identified and remediated security vulnerabilities flagged through code reviews and scanning tools (Black Duck, Checkmarx, SonarQube), applying secure coding practices in collaboration with the security engineering team
- Created an internal dashboard to track and visualise vulnerabilities across scanning tools, increasing visibility for engineering teams and helping reduce remediation turnaround time

Software Engineer Intern – Visa, Bangalore, India (May 2021 – Jul 2021)

- Set up integration testing using TestContainers, improving reliability and consistency of test environments
- Migrated the testing setup from an in-memory H2 database to containerised MySQL, aligning test environments more closely with production for more accurate results

Software Validation Engineer – Broadcom, Bangalore, India (Jun 2018 – Nov 2018)

- Automated WLAN test cases using Python within an internal test automation framework, reducing manual testing effort and enhancing test execution efficiency

PROJECTS

Real-Time Chat Application – Node.js, React, Express.js, MongoDB, Socket.IO – [GitHub](#)

- Architected a real-time messaging platform using WebSockets (Socket.IO), supporting low-latency communication between concurrent users
- Designed a full-stack, event-driven architecture combining WebSockets and REST APIs, with a responsive React-based UI

Swift Typer Application – React, Tailwind CSS – [GitHub](#) – [Live Demo](#)

- Built an interactive typing speed application with real-time performance tracking and theme-switching functionality
- Implemented a responsive, component-based UI with smooth animations for an enhanced user experience

Dynamic Graph Algorithms for GPU – CUDA, C++ – [GitHub](#)

- Engineered GPU-accelerated algorithms for dynamic graph processing as part of the Starplat DSL research project, enhancing computational efficiency for large-scale graph operations

EDUCATION

- **MSc Cyber Security** – University of Chester (Jan 2025 – Feb 2026) – Distinction
- **MTech Computer Science and Engineering** – IIT Madras (Aug 2020 – Jun 2022) – CGPA: 9.44
- **BTech Computer Science and Engineering** – CUSAT (Aug 2014 – Jun 2018) – CGPA: 7.47

ACHIEVEMENTS & LEADERSHIP

- **Student Voice Representative, University of Chester (Sep 2025 – Feb 2026)** – Represented student feedback to faculty, contributing to academic and welfare improvements
- **Graph DSL Patent Contributor, IIT Madras (Jul 2021 – Jun 2022)** – Contributed to development of a patented Domain-Specific Language for graph processing
- **Star Teaching Assistant, IIT Madras (Jul 2021 – Jun 2022)** – Recognised for exceptional academic support in Computer Science courses
- **Placement Coordinator, IIT Madras (Jul 2021 – Jun 2022)** – Coordinated recruitment activities with 480+ companies
- **Prakash Arora Prize, IIT Madras (Aug 2020 – Jun 2021)** – Awarded for highest CGPA in first year of MTech
- **GATE CSE Rank 591 (Mar 2020)** – Secured All India Rank 591 among ~100,000 candidates